Why Laya collapses on 72 options: a working prefix strip helps, and position and a distinct opening both pull the collapse - an exploratory follow-up

PinoAI research note · THEMIS-BENCH-002 · exploratory, one benchmark, n = 100

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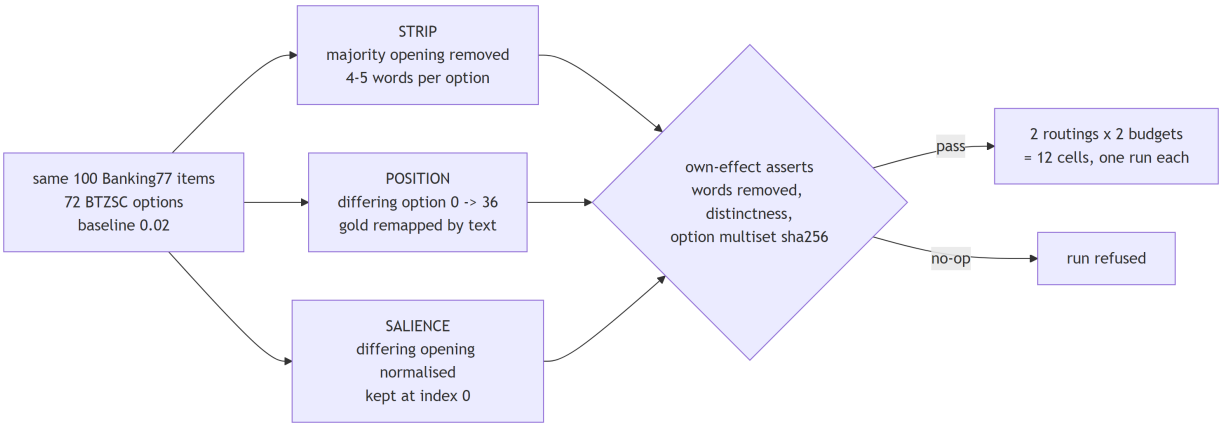
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Abstract

THEMIS-BENCH-001 found that option wording, more than the option budget, drives Laya's collapse on BTZSC's 72-option Banking77 (0.02, at chance), but it left two loose ends: its "prefix stripped" rows removed one word (PM-023), and the forced-multilingual route put 98 of 100 predictions on option 0 - the first position and the only option whose opening differs. This note follows up on the same 100 items with three treatments, each of which asserted its own effect before any model call.

1. **A working prefix strip helps.** Removing the majority opening ("This banking customer example message", 4-5 words from every option) raises accuracy from 0.02 to 0.11 (default routing, p = 0.022), 0.12 (forced multilingual, p = 0.013), and 0.24 with a 1,024-token budget on the multilingual route (p = 6.1e-05 against the same-budget baseline). Paired McNemar exact tests, same items, one run per cell.
2. **It does not close the wording gap.** The best strip cell (0.24) stays below the 77 PolyAI category names on the same items (0.37; 23 vs 10 discordant, p = 0.035). As in BENCH-001, that contrast changes wording and label set together.
3. **The two pre-declared cells point to different causes; no mechanism is claimed.** Under the readings declared before the run, the position cell reads SALIENCE (moving the differing option to index 36 moves 99/100 predictions with it) and the salience cell reads POSITION (normalising its opening still leaves 67/100 on index 0). The cells disagree, so by the pre-declared rule we report both and claim nothing about a single cause.

1. Design



Three treatments, each proving its own effect before any model call; readings declared before the run.

Laya 0.3.28 at the registered pins, CPU fp32, offline, under saferun; progress records every 25 items. Every treatment asserted its own effect on a dry run before any model call (ml-engineer 10bca6): the strip removes 4-5 words from every option and leaves all 72 distinct; the position control changes only the index of the differing option (option-multiset sha256 unchanged, gold remapped and asserted by text); the salience control changes only that option's opening. A first

draft of the strip rule ("up to the first 'is'") was refused by its own assert, because one option reads "… message requests ATM support." - the process change from PM-023 working before any number existed.

Readings declared before any number (data-science ebc0a1; registered): on the forced-multilingual route, the POSITION cell reads SALIENCE if at least 50% of predictions follow the moved option to index 36 and POSITION if at least 50% stay on index 0; the SALIENCE cell reads POSITION if the collapse on index 0 persists (at least 50%) and SALIENCE if it disappears. Both cells must agree for any mechanism claim; if they disagree, both are reported and nothing is claimed. Accuracy deltas are paired McNemar exact tests against the same-routing, same-budget BTZSC baseline, per cell, with no family-wise claim.

2. Results

Table 1: Twelve exploratory cells on the same 100 items, one run each. Baselines: 0.02 (default and multilingual, Laya's budget), 0.09 (multilingual, 1,024). Default routing at 1,024 errors on every item because the English checkpoint's max_len is 512 (as in BENCH-001).

Treatment	Routing	Budget	Accuracy (Wilson 95%)	vs same baseline (cell-only / baseline-only)	McNemar exact p
strip	default	Laya default	0.11 [0.06, 0.19]	11 / 2	0.022
strip	multilingual	Laya default	0.12 [0.07, 0.20]	12 / 2	0.013
strip	multilingual	1,024	0.24 [0.17, 0.33]	15 / 0	6.1e-05
position	default	Laya default	0.03	2 / 1	1
position	multilingual	Laya default	0.02	0 / 0	1
position	multilingual	1,024	0.08	0 / 1	1
salience	default	Laya default	0.02	2 / 2	1
salience	multilingual	Laya default	0.03	1 / 0	1
salience	multilingual	1,024	0.08	0 / 1	1
all three	default	1,024	100 runtime errors each	-	-

Table 2: The mechanism cells. At Laya's budget the two cells disagree; at 1,024 they disagree too. By the pre-declared rule no single mechanism is claimed.

Forced multilingual	Predictions on index 0	on index 36	Pre-declared reading
POSITION cell, Laya budget (differing option at 36)	0	99	SALIENCE
SALIENCE cell, Laya budget (opening normalised, at 0)	67	0	POSITION
POSITION cell, 1,024	1	3	unresolved
SALIENCE cell, 1,024	2	0	SALIENCE

3. Reading

What the strip shows. Once the majority opening is actually removed, Laya leaves chance on every route that scores, and the gain grows with budget (0.24 at 1,024 on the multilingual route). So the shared opening does cost Laya accuracy - the hypothesis BENCH-001 could not test. But the best strip cell is still 13 points below the category-name row on the same items ($p = 0.035$), so wording/label set remains the largest lever we have measured.

What the mechanism cells show, descriptively. At Laya's budget on the multilingual route, a distinct opening captures the collapse wherever it sits (99/100 at index 36), and when no option is distinct, the collapse falls back to the first position (67/100 at index 0). Both readings are true of different cells, which is exactly what the pre-declared rule exists for: we report both and claim neither as "the" mechanism. A hypothesis for later work, not a claim: a distinct opening dominates when present, with a residual first-position pull when absent.

Further observations, flagged and not claimed. At 1,024 on the multilingual route, both mechanism cells collapse onto the same two other indices (59 and 64; 36-37 and 22-23 of 100), an attractor independent of the differing option (data-science 89947e). On the default route, both mechanism cells collapse onto index 1 (75/100 and 56/100). Neither was declared in advance; they are observations for the next design, not findings.

4. What this changes for Themis

- The Themis-vs-Laya comparison keeps BENCH-001's rule: identical option text and framing on both sides. This note adds that Laya's many-option score is also sensitive to option ORDER and to one odd option, so the comparison fixes option order per item and reports Laya under the released order.
- Themis design target W1 (option-wise scoring with full context) is tested against exactly these failure modes: label-permutation consistency (W5) and a single-odd-option control join the Themis evaluation suite.

5. Limitations

- Exploratory: 12 cells, one run each, one benchmark, n = 100; per-cell p-values with no family-wise correction. The only significant results are the three strip cells; the mechanism cells did not change accuracy (all p = 1).
- The mechanism readings disagree, so no mechanism is claimed. The 59/64 and index-1 attractors were not declared in advance.
- The strip-versus-names contrast changes wording and label set together (72 BTZSC hypotheses vs 77 PolyAI categories), as in BENCH-001.
- Default routing at 1,024 is unmeasured (errors), as before.

Reproducibility

Table 3: Files.

Item	Where
Fix-run report	`experiments/results/themis/laya-bench/mech/report-b77-mech.json` (sha256 e84b6f67; git-ignored, rendered in /themis)
Script	`scripts/judge/laya_bench_b77_mech.py` (708ad0e4)
Numbers in this note	`experiments/reports/_themis_bench002.py` asserts each one from the preds files. Its first draft had a paired-count helper that compared one row's prediction with the other row's target - wrong across label spaces; the checker failed on it (36 / 22 instead of 23 / 10), the helper now scores each row against its own target, and BENCH-001's identical helper was re-derived both ways (37 / 2, unaffected) and fixed
Pre-declared readings	data-science ebc0a1, registered in the DECISION_LOG before any number

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AI assistance. Analysis code, statistical checks and drafts of this document were prepared with the help of Claude (Anthropic), an AI model. Every number is generated from the project's result files. The author reviewed the content and takes responsibility for it.